

Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

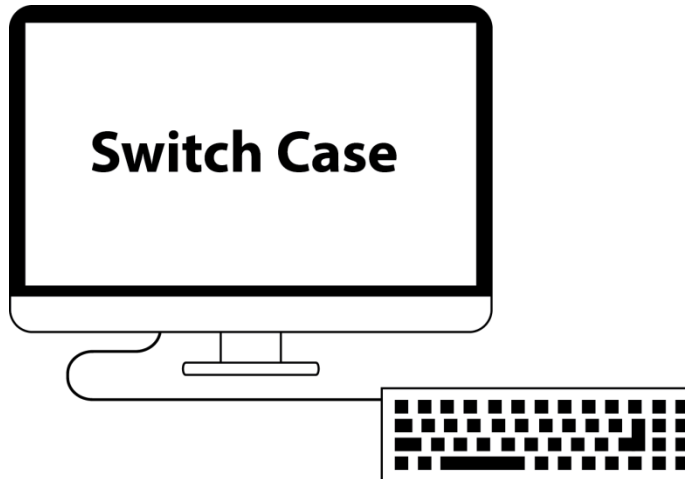
### CONTENT

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### Let's Start!

#### Considering Multiple Conditions



#### Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

#### Can Do Statements:

- I can switch statements in JavaScript.
- I can identify information in a flowchart of a switch statement.
- I can talk about the syntax to use in a switch statement in JavaScript.

#### Ask your partner:

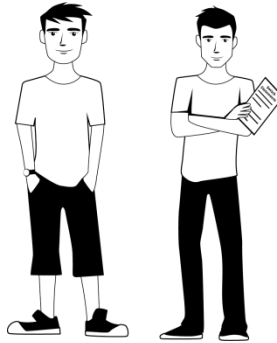
- How do you think conditions in Javascript work?
- Why do you think conditions are important when programming?





### Let's Explore!

Elsa and Glenn review the switch statement topic to check if they understood correctly.



*Instructions: Read the text below.*

**Glenn:** Hey Elsa! Could you help me understand what switch statements are?

**Elsa:** Sure! I was actually studying for our JavaScript class.

**Glenn:** Great! I know we can use switch statements instead of the 'if...else...if' ones when we have to deal with different conditions that depend on the value of a single variable, right?

**Elsa:** That's right. When we want to perform a multiway branch, we may want to use switch statements. This could simplify the decision making process. Basically, the objective of switch statements is to evaluate several statements based on the value of an expression. The program uses an interpreter which goes over each case and compares them with the expression until a match is found. If the interpreter goes over all the cases and finds no match, the default condition will be executed.

**Glenn:** Do you think that knowing the syntax may help me understand how switch statements work more easily?

**Elsa:** It may. Let me tell you how syntax works. First, we have to write 'switch (expression)'. Here, you will write the value of the variable that you are going to be working with. Then, you are going to write the different conditions.

**Glenn:** I think this process may be difficult. There are many conditions to take into account.

**Elsa:** You can write a list on a piece of paper and then write the code. If you realize you missed a condition, you could edit the code. Make sure you write 'break' after you've indicated each condition. This will show where each case ends. Remember that you are going to write the conditions starting with 'case', then you'll assign a letter or a number to each case. After this, you will need to indicate the conditions inside a parenthesis. Once you've done this, you'll write a default statement that will be executed if none of the previous conditions matched.

**Glenn:** If my program shows a default statement, could I have missed a condition?

**Elsa:** Maybe, but you will have to go through the code to make sure you didn't miss anything. I want to show you a flowchart that will clarify how switch statements work. As you can see, we first have the expression. Then, the interpreter goes case by case looking for a matching statement. If it finds a matching statement, it'll execute it; if not, the interpreter will evaluate the following statement. This process will go on until the program has found either a matching statement or the default block.

**Glenn:** Thank you, Elsa! Everything is much clearer now.





### Let's Practice!

#### Using Switch Statement is Very Handy

*Instructions: Read the text and select all the subject and object relative clauses. Check your answers with your partner.*

Programmers that work with JavaScript often use "if... else... if" statements to perform a multiway branch, but sometimes doing that isn't very handy. We can skip the repetition of "if... else... if" statements which depend on a single variable by using Switch statements. You can include different statements which are evaluated and executed by one expression. Doing that saves a lot of time because during the interpretation that evaluates the different combinations the program finds the appropriate match, and if a match isn't found, which is sometimes possible, the program executes a default condition.



### Let's Practice!

#### Differentiating Expressions from Conditional Statements

*Instructions: Classify the sentences by writing the number of the sentence in the correct category. Check your exercise with your partner. Discuss.*

1. Switch Day
2. Case Doubt: `println ("You could leave the difficult questions for last")`
3. Switch JavaScript
4. Case Monday: `println ("You could finish your JavaScript manual")`
5. Switch Exam
6. Case Variables: `println ("You may need to define the variables first")`

| Examples of Expressions | Examples of Conditional Statements |
|-------------------------|------------------------------------|
|                         |                                    |



### Extension Activity

Design an innovative loan system for your community.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Search for more information about Switch Statements in JavaScript.
- Create a chart with examples comparing Switch Statements and "if... else... if" statements.
- Prepare a poster including what Switch statements are and an example.
- Share the poster with the class.

#### Self-Assessment

Yes

No

- I use switch statements.
- I identify information in a flowchart of a switch statement.
- I understand the syntax for a switch statement.
- I share my work with the class.

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### Target Words

- multiway** (adjective): having different possibilities or paths to take  
e.g. When we want to perform a multiway branch, we may want to use switch statements.
- branch** (noun): a part or section of a structure  
e.g. When we want to perform a multiway branch, we may want to use switch statements.
- execute a condition** (expression): to choose and process a given condition to carry out a specific function in a program  
e.g. If the interpreter goes over all the cases and finds no match, the default condition will be executed.
- match** (verb): to correspond or have compatible characteristics with a different element  
e.g. Once you've done this, you'll to write a default statement that will be executed if none of the previous conditions matched.
- default** (adjective): having a value that the program assumes when no other values were specified  
e.g. If the interpreter goes over all the cases and finds no match, the default condition will be executed.



## Learning Points

### Modal Verbs for Speculating: May and Could

There are **modal verbs** used to speculate about something.

We use **modal verbs** to express obligation, possibility, certainty and speculation, among others. In this learning point, we are going to talk about the modal verbs '**may**' and '**could**' that can be used to speculate or express an assumption about a given situation.

We use **could** to indicate that a situation or result is highly probable. On the other hand, **may** is used to express that a situation or result is slightly possible.

Examples:

*You **could** use conditional statements in Javascript to help your program make correct decisions.*

*You **may** use the 'if...else...if' statement instead of the 'if...else' one for this specific situation. But you'd better double check with the manager.*

Finally, remember that we can use **modal verbs** in different tenses and forms depending on what we want to express.

Examples:

*This code **may have been written** by the software development team.*

***Could** we use switch statements in Javascript?*

*He **may not** have been ready for this company. I think that's why he quitted.*

### Examples

- *When we want to perform a multiway branch, we may want to use switch statements. This could simplify the decision making process.*
- *Do you think that knowing the syntax may help understand how switch statements work more easily?*
- *I think this process may be difficult. There are many conditions to take into account.*

### Flowchart of How a Switch Statement Works

**Switch statements** integrate multiple "if...else...if" statements.

When we have different conditions, we can use multiple 'if...else .. if' statements for performing a multiway branch, Nevertheless, we can also use switch statements to accomplish the same goal in a more efficient way, especially if all the branches depend on the value of one variable. Let's take a look at the following flowchart:

